torch.cuda.cudart#

https://pytorch.org/docs/stable/generated/torch.cuda.cudart.html#torch.cuda.

Description:

Retrieves the CUDA runtime API module. This function initializes the CUDA runtime environment if it is not already initialized and returns the CUDA runtime API module (_cudart). The CUDA runtime API module provides access to various CUDA runtime functions. The CUDA runtime API module (_cudart).

Function Signature

```
torch.cuda.cudart()[source]#

Parameters

Returns

Return type

Raises
```

Returns:

module

Code Examples

```
>>> import torch
>>> from torch.cuda import cudart, check_error
>>> import os
>>>
>>> os.environ["CUDA_PROFILE"] = "1"
>>>
>>> def perform_cuda_operations_with_streams():
>>>     stream = torch.cuda.Stream()
>>>     with torch.cuda.stream(stream):
>>>         x = torch.randn(100, 100, device='cuda')
>>>         y = torch.randn(100, 100, device='cuda')
>>>         z = torch.mul(x, y)
>>>     return z
>>>
>>> torch.cuda.synchronize()
>>> print("====== Start nsys profiling ======")
>>> check_error(cudart().cudaProfilerStart())
>>> with torch.autograd.profiler.emit_nvtx():
>>>     result = perform_cuda_operations_with_streams()
>>>     print("CUDA operations completed.")
>>> check_error(torch.cuda.cudart().cudaProfilerStop())
>>> print("====== End nsys profiling ======")
```

```
>>> $ nvprof --profile-from-start off --csv --print-summary -o
trace_name.prof -f -- python cudart_test.py
```

Detailed Documentation

Documentation

Retrieves the CUDA runtime API module.

This function initializes the CUDA runtime environment if it is not already initialized and returns the CUDA runtime API module (_cudart). The CUDA runtime API module provides access to various CUDA runtime functions.

The CUDA runtime API module (_cudart).

RuntimeError – If CUDA cannot be re-initialized in a forked subprocess.

AssertionError – If PyTorch is not compiled with CUDA support or if libcudart functions are unavailable.

This command profiles the CUDA operations in the provided script and saves the profiling information to a file named trace_name.prof. The –profile-from-start off option ensures that profiling starts only after the cudaProfilerStart call in the script. The –csv and –print-summary options format the profiling output as a CSV file and print a summary, respectively. The -o option specifies the output file name, and the -f option forces the overwrite of the output file if it already exists.